# Accessibility on Demand™

Product Requirements Document for Enterprise PDF Accessibility Platform

| AUTHOR | STATUS | UPDATED | VERSION |
|---|---|---|---|
| Poulastha Mukherjee | Production Ready | 24.07.2025 | 2.0 |

## Problem Statement

**Organizations face massive compliance risks and costs when trying to make PDF documents accessible at scale, with traditional manual remediation costing $8-$50+ per page and taking 15-30+ minutes per page with 30% error rates.**

The current PDF accessibility landscape presents critical challenges:

- **Regulatory Pressure:** USA's April 2026/27 ADA Title II deadlines approaching with legal cases averaging $100,000+
- **Scale Impossibility:** Manual remediation cannot handle enterprise document backlogs of thousands or millions of PDFs
- **Cost Prohibitive:** Traditional accessibility consulting ranges from $8-$50+ per page, making large-scale compliance economically unfeasible
- **Quality Inconsistency:** 30% error rate in manual remediation with unpredictable compliance check results
- **Complex Standards:** Multiple overlapping standards (WCAG 2.1 AA, PDF/UA, Section 508, EN 301 549) create confusion
- **Resource Intensive:** Organizations lack specialized accessibility expertise internally

This creates an urgent need for enterprises, government agencies, and educational institutions to find scalable, reliable, and cost-effective solutions before regulatory deadlines.

## Market Opportunity

**Transform the $2.8B+ document accessibility market by providing enterprise-grade AI-powered PDF remediation that delivers 95%+ compliance scores at 90% cost reduction compared to traditional methods.**

The market drivers are compelling:

- **Regulatory Timeline:** ADA Title II/III compliance deadlines create immediate demand
- **Global Standards Alignment:** International accessibility laws all reference WCAG/PDF/UA standards
- **Enterprise Backlogs:** Organizations have millions of legacy PDFs requiring remediation
- **Risk Mitigation:** Legal exposure averaging $100,000+ per accessibility lawsuit
- **Digital Transformation:** Increased focus on inclusive digital experiences

WCAG 2.1 AA | PDF/UA | Section 508 | EN 301 549 | AODA | DDA | RGAA | BITV 2.0

### Core Value Proposition

"The first AI-powered enterprise platform that automatically remediates PDF documents at scale, delivering 95%+ compliance scores against global accessibility standards in minutes instead of months, at 90% cost reduction."

## Target Users & Use Cases

### Primary: Enterprise Organizations

**Who:** Large corporations, government agencies, educational institutions

**Volume:** 10,000+ PDFs, ongoing document production

**Pain:** "We have millions of legacy documents and tight compliance deadlines with no scalable solution"

**Goal:** Achieve compliance at scale without breaking budget or missing deadlines

### Secondary: Mid-Market Organizations

**Who:** Regional businesses, smaller government entities, nonprofits

**Volume:** 1,000-10,000 PDFs, periodic batch processing

**Pain:** "Traditional accessibility services are too expensive and slow for our needs"

**Goal:** Cost-effective compliance solution with reliable quality

### Tertiary: SMB & Individual Users

**Who:** Small businesses, consultants, individual professionals

**Volume:** <1,000 PDFs, on-demand processing

**Pain:** "Need occasional accessibility remediation without service contracts"

**Goal:** Simple, transparent, pay-per-use accessibility solution

## Solution Architecture

**An enterprise-grade, AI-powered SaaS platform that automatically processes any PDF format (scanned, digital, complex layouts) and delivers compliance-ready documents with third-party validation.**

### Core Processing Pipeline

Upload → AI Analysis → Auto-Remediation → Compliance Check → Validation Report

### Platform Architecture

**Frontend Layer:** React + TypeScript + Mantine UI + Redux Toolkit

**API Gateway:** RESTful APIs + Socket.IO for Real-time Updates

**AI Processing Engine:** Proprietary OCR + Vision AI + NLP Pipeline

**Compliance Validation:** Third-party PAC Checker Integration

**Storage & Security:** AWS S3 + Encryption + Audit Logging

## Technology Stack

**Frontend Framework**
- React 18.2.0
- TypeScript
- Vite Build Tool

**UI Components**
- Mantine Component Library
- Tailwind CSS

**State Management**
- Redux Toolkit
- React Query (TanStack)
- React Router

**PDF Processing**
- React-PDF Renderer
- jsPDF
- Custom OCR Engine

**Backend Services**
- FastAPI
- Node.js + Express
- Socket.IO
- AWS Lambda Functions

**Infrastructure**
- AWS S3 Storage
- Database (PostgreSQL)
- Redis Caching

## Service Tiers & Pricing

### Standard

**$0.25/page**

*Best for text-heavy documents & internal archives*

- ✓ Searchable Text Layer Creation
- ✓ Automated Tag Generation (H, P, L)
- ✓ Logical Reading Order
- ✓ Basic Alt Text (Image Type Tagging)
- ✓ Compliance Report & Quality Check

### Enhanced

**$1.50/page**

*95%+ compliance score guarantee*

- ✓ Everything in Standard plus:
- ✓ Contextual Alt Text (Images/Charts)
- ✓ Advanced Form Tagging
- ✓ Configurable Bookmarks
- ✓ 95%+ Compliance Score Guarantee
- ✓ Third-party PAC Validation

### Full Remediation

**$10+/page**

*Human-verified 100% compliance*

- ✓ Everything in Enhanced plus:
- ✓ Expert Human Review & Correction
- ✓ 100% Compliance Score Guarantee
- ✓ Full WCAG, PDF/UA Certification
- ✓ Fast Turnaround SLAs
- ✓ Legal-ready Documentation

## Key Features & Requirements

### Core Platform Features

| PRIORITY | FEATURE | DESCRIPTION | USER IMPACT |
|---|---|---|---|
| P0 | Batch Upload & Processing | Drag & drop or bulk upload with parallel processing | Handle enterprise volume efficiently |
| P0 | AI-Powered Auto-Remediation | Proprietary vision AI handles any document format | Consistent quality without human intervention |
| P0 | Real-time Progress Tracking | Live updates on processing status via Socket.IO | Transparency and confidence in processing |
| P0 | Third-party Compliance Validation | Independent PAC Checker integration for scores | Audit-ready compliance documentation |
| P0 | Interactive Review Interface with PDF Metadata | Tag tree editing, reading order adjustment | Quality control and customization capability |

### Enterprise Integration Features

| PRIORITY | FEATURE | DESCRIPTION | BUSINESS VALUE |
|---|---|---|---|
| P0 | API Integration | RESTful APIs for IDP stack integration | Seamless workflow automation |
| P0 | AWS Integration | Direct S3 bucket access and processing | Enterprise security and storage compatibility |
| P1 | Volume Discounting | Automatic pricing tiers for large volumes | Cost optimization for enterprise clients |
| P1 | Dedicated Account Management | Assigned support for enterprise customers | White-glove service experience |
| P1 | Audit Trail & Reporting | Comprehensive compliance documentation | Legal and regulatory requirement fulfillment |

### User Experience Requirements

| PRIORITY | REQUIREMENT | SUCCESS CRITERIA |
|---|---|---|
| P0 | Intuitive Interface | Zero training required, self-service capability |
| P0 | Processing Speed | Standard/Enhanced: Minutes, Full: 24-48 hours |
| P0 | Quality Assurance | 95%+ compliance scores with guarantees |
| P1 | Multi-format Support | Scanned, digital, forms, complex layouts |
| P1 | Scalability | Handle thousands of documents simultaneously |

## Success Metrics & KPIs

| CATEGORY | METRIC | TARGET | MEASUREMENT METHOD |
|---|---|---|---|
| **Quality** | Compliance Score Achievement | 95%+ for Enhanced, 100% for Full Remediation | Third-party PAC Checker validation |
| **Performance** | Processing Speed | Standard/Enhanced: <5 minutes/document | System performance monitoring |
| **Scale** | Monthly Page Volume | >1M pages processed/month | Platform analytics dashboard |
| **Reliability** | System Uptime | 99.9% availability | Infrastructure monitoring |
| **Customer Satisfaction** | Net Promoter Score | NPS >60 | Customer surveys and feedback |
| **Business Growth** | Enterprise Client Retention | >90% annual retention | Account management tracking |
| **Cost Efficiency** | Cost per Page vs Traditional | 90% cost reduction vs manual | Market pricing analysis |

## User Workflows & Journey

### Enterprise User Journey

**1. Onboarding & Setup**

Account Creation → AWS Integration → API Setup → Training Session

**2. Batch Processing Workflow**

Create Document Set → Bulk Upload → Service Level Selection → Processing Monitor → Quality Review → Download Results

### Key User Interactions

| USER ACTION | SYSTEM RESPONSE | EXPECTED OUTCOME | PERFORMANCE TARGET |
|---|---|---|---|
| Upload PDF batch | Real-time upload progress, file validation | All files queued for processing | <10MB/sec upload speed |
| Select service level | Cost calculation, processing time estimate | Clear pricing transparency | Instant calculation |
| Monitor processing | Live status updates via WebSocket | Real-time visibility into progress | Updates every 5 seconds |
| Review results | Interactive compliance report, tag editor | Quality validation and editing capability | Report loads in <3 seconds |
| Download files | Batch download with compliance certificates | Complete deliverable package | Download ready instantly |

## Compliance & Standards Coverage

**AoD™ ensures compliance with all major global accessibility standards by targeting WCAG 2.1 AA and PDF/UA as the foundational requirements referenced by international regulations.**

### Global Standards Mapping

| REGION/COUNTRY | STANDARD/LAW | CORE REQUIREMENTS | AOD™ COVERAGE |
|---|---|---|---|
| **United States** | ADA Title II/III, Section 508 | WCAG 2.1 AA compliance | ✅ Full Coverage |
| **European Union** | EN 301 549 | WCAG 2.1 AA + PDF/UA | ✅ Full Coverage |
| **Canada** | AODA | WCAG 2.0/2.1 AA | ✅ Full Coverage |
| **Australia** | DDA | WCAG 2.1 AA | ✅ Full Coverage |
| **United Kingdom** | PSBAR | WCAG 2.1 AA | ✅ Full Coverage |
| **France** | RGAA | WCAG 2.1 AA | ✅ Full Coverage |
| **Germany** | BITV 2.0 | WCAG 2.1 AA | ✅ Full Coverage |

### Compliance Score Calculation

#### Industry-Standard Validation

**Compliance Score % = (Total Checks - Failed Checks) / Total Checks × 100%**

Scores are independently calculated using the industry-standard PAC (PDF Accessibility Checker), evaluating hundreds of validation points to ensure comprehensive compliance. This objective methodology removes subjectivity and provides audit-ready documentation.

**🎯 Standard Tier**

Basic compliance coverage focusing on structural accessibility and searchability for internal documents.

**🏆 Enhanced Tier**

95%+ compliance score guarantee with comprehensive WCAG 2.1 AA and PDF/UA validation for public-facing documents.

**💎 Full Remediation**

100% compliance score guarantee with human expert review for legally mandated documents requiring absolute compliance.

## Technical Architecture Deep Dive

### AI Processing Pipeline

#### Document Analysis & Remediation Engine

**Input Processing:** Multi-format PDF ingestion with metadata extraction

**Vision AI Layer:** Advanced OCR with image cleanup and text recognition

**Structure Analysis:** Automated tagging engine for headings, paragraphs, lists

**Content Enhancement:** Contextual alt-text generation and form tagging

**Compliance Validation:** Real-time checking against WCAG/PDF-UA standards

**Output Generation:** Accessible PDF creation with compliance certification

### Platform Infrastructure

| COMPONENT | TECHNOLOGY | PURPOSE | SCALABILITY |
|---|---|---|---|
| **Frontend Application** | React + TypeScript + Mantine | User interface and experience | CDN distribution, lazy loading |
| **API Gateway** | FastAPI + Node.js + Express | Request routing and authentication | Load balancing, auto-scaling |
| **Processing Engine** | AWS Lambda + Custom AI | PDF analysis and remediation | Serverless parallel processing |
| **File Storage** | AWS S3 + CloudFront | Document storage and delivery | Global distribution, versioning |
| **Database** | PostgreSQL + Redis | User data and processing state | Read replicas, caching |
| **Real-time Updates** | Socket.IO | Live processing status | WebSocket clustering |

### Security & Compliance

**🔒 Data Encryption**

End-to-end encryption for all document uploads and processing, with AES-256 encryption at rest and TLS 1.3 in transit.

**🛡️ Access Control**

Role-based access control with JWT authentication, session management, and API key security for enterprise integrations.

**📋 Audit Logging**

Comprehensive audit trails for all document processing activities, user actions, and system events for compliance reporting.

**🏢 Enterprise Security**

SOC 2 Type II compliance, GDPR/CCPA adherence, and enterprise SSO integration capabilities.

## Competitive Advantages

**AoD™ is the only enterprise platform that combines AI-powered automation, third-party validation, and transparent pricing to deliver institutional-grade accessibility at consumer-friendly costs.**

### Competitive Comparison

| FACTOR | TRADITIONAL SERVICES | OTHER TOOLS | AOD™ ADVANTAGE |
|---|---|---|---|
| **Cost per Page** | $8-$50+ | $2-$10 | $0.25-$1.50 (90% savings) |
| **Processing Time** | 15-30 minutes/page | 5-15 minutes/page | Seconds to minutes (95% faster) |
| **Quality Consistency** | 70% accuracy (manual errors) | 80-85% accuracy | 95%+ guaranteed compliance |
| **Scale Capability** | Limited by human capacity | Moderate automation | Unlimited parallel processing |
| **Transparency** | Black box process | Limited visibility | Real-time progress + validation |
| **Enterprise Integration** | Manual handoffs | Basic APIs | Full IDP stack integration |

### Unique Value Propositions

**🚀 Speed at Scale**

Process thousands of documents simultaneously with AI-powered parallel processing that traditional services cannot match.

**🎯 Guaranteed Quality**

95%+ compliance scores backed by third-party validation and money-back guarantees remove quality risk.

**💡 Transparent Process**

Real-time visibility into processing status and detailed compliance reports eliminate the "black box" problem.

**🔌 Enterprise Ready**

API-first architecture with AWS integration and enterprise security makes adoption seamless for large organizations.

## Implementation Roadmap

### Development Phases

| PHASE | TIMELINE | KEY FEATURES | SUCCESS CRITERIA |
|---|---|---|---|
| **Phase 1: Core Platform** | Q1 2025 ✅ | Basic upload, AI processing, compliance scoring | 500+ documents processed daily |
| **Phase 2: Enterprise Features** | Q2 2025 | API integration, batch processing, AWS connectivity | 10+ enterprise clients onboarded |
| **Phase 3: Advanced AI** | Q3 2025 | Enhanced contextual analysis, form optimization | 98%+ compliance scores achieved |
| **Phase 4: Global Scale** | Q4 2025 | Multi-region deployment, advanced analytics | 1M+ pages processed monthly |

### Go-to-Market Strategy

**Market Entry Approach**

Enterprise Pilots → Industry Validation → Partner Channel → Market Expansion

## Risk Assessment & Mitigation

| RISK CATEGORY | POTENTIAL IMPACT | PROBABILITY | MITIGATION STRATEGY |
|---|---|---|---|
| **AI Quality Variance** | Compliance scores below guarantee | Medium | Continuous model training, human review fallback, insurance coverage |
| **Regulatory Changes** | Standards evolution affecting compliance | High | Continuous monitoring, flexible architecture, rapid deployment capability |
| **Scale Performance** | System degradation under high load | Medium | Auto-scaling infrastructure, load testing, performance monitoring |
| **Competitive Response** | Market pricing pressure | High | Continuous innovation, customer lock-in via integration, patent protection |
| **Security Breach** | Data compromise, regulatory penalties | Low | Enterprise security standards, regular audits, cyber insurance |

## Appendix

### Technical Specifications

- **Supported File Types:** PDF (all versions), scanned documents, digital PDFs, fillable forms
- **File Size Limits:** Up to 500MB per file, unlimited batch sizes
- **Processing Capacity:** 10,000+ concurrent documents
- **API Rate Limits:** 1000 requests/hour (standard), unlimited (enterprise)
- **Data Retention:** 30 days (standard), custom (enterprise)
- **Compliance Certifications:** SOC 2 Type II, GDPR, CCPA, HIPAA (available)

### Integration Examples

**Document Management**

SharePoint, Box, Dropbox, Google Drive integration for automated processing workflows

**Content Management**

Drupal, WordPress, Sitecore integration for web content accessibility

**Enterprise Systems**

SAP, Oracle, Microsoft 365 integration for business document processing

### Support & Training

- **Documentation:** Comprehensive API docs, user guides, best practices
- **Training:** Platform onboarding, administrator training, developer workshops
- **Support Tiers:** Community (free), Business (email), Enterprise (dedicated)
- **SLA Guarantees:** 99.9% uptime, <4 hour response (enterprise)